08C — Deep-Dive Feature Audit: Outside-Sect World & Ecology (Part 3 of 6)

Secret Realms, 43-Floor Nine-Dao Pagoda, 5-Elemental Realms, 4 Ancient Sanctums, Elemental Puzzle Gates, 4 Micro-Locations & 3-Layer Immortal Tombs

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Dungeons, towers, sanctums, ancient tombs, and elemental ruins must be conquered by real autonomous parties marching out with complementary paths, elements, and skills—never capped at legacy Floor 9, never conjuring stones out of an empty world pool, and never clearing 3-layer Emperor Tombs with `0` party members!
- **Master Roadmap for Category 08 (`08A–08F`)**:
  - `08A`: Macro Geopolitics, 19 Rival Sects, Diplomacy, Vassals, 6 Resource Nodes & Siege Warfare
  - `08B`: Expeditions, Caravans, Midnight Black Market, Tavern Bounties, Espionage & `v2.6.0` Living Chronicle
  - **`08C` (This Document)**: **Secret Realms (`OUT-18`), 43-Floor Nine-Dao Pagoda, 5-Elemental Realms, 4 Sanctums, Puzzle Gates, 4 Micro-Locations (`OUT-05`) & 3-Layer Immortal Tombs (`explore.js`, `sect.js`, `expedition.js`)**
  - `08D`: Wilderness Ecology, Roaming Beast Packs, Taming/Pets/Breeding/Menagerie/Symbiosis & Wild Resource Gathering
  - `08E`: Southern Spirit River Fishing, Spirit Cuisine, Underwater Chests, Wandering Figures, Visitors & Legendary Clue Quests
  - `08F`: Grand Finale — Holistic Overview of the Entire Outside-Sect World & Ecology

---

1. `[REALM-PAG-01]` Critical Reachability Lockout: `34` of the `43` Nine-Dao Pagoda Floors (`Floors 10–43`) & the 4th Ancient Sanctum Have `0.00%` Autonomous Visitation (`js/explore.js:738–742`)!

1. **Hardcoded `{ maxFloor: 9 }` in `EX.tryVolunteer` (`js/explore.js:742`)**:

- Patch `v2.3.0` expanded `NINE_DAO_PAGODA` to **43 Floors** (`1` floor per Major Path, dropping `RES_NINE_DAO_<PATH>` set pieces and Grand Dao scriptures).
- **Live Bug**: `EX.tryVolunteer` (`line 742`) still passes `{ maxFloor: 9 }` from the 9-path era! Autonomous NPC parties **never enter Floors `10–43`**, locking **79.1% (`34/43`) of Pagoda floors, set pieces (`05D`), and path manuals** out of natural gameplay!
- **Fix**: Compute `const maxFloor = clamp(Math.round(partyAvgEquivStage * 5), 5, 43)` (or `Math.min(43, (W.explore.pagoda?.highestCleared || 0) + 4)`).

2. **`SANCTUM_MYRIAD_BEAST_TOTEM` Completely Omitted From `EX.tryVolunteer` (`js/explore.js:738`)**:
   - Line `738` selects only `'SANCTUM_BLOOD_VAJRA'` (`BODY`), `'SANCTUM_CELESTIAL_ORACLE'` (`GOD`), `'SANCTUM_ABYSSAL_SOUL'` (`DEMON`), and defaults all other 40 paths to `'SANCTUM_SWORD_ABYSS'`—**never selecting the 4th Sanctum `'SANCTUM_MYRIAD_BEAST_TOTEM'`**!
   - **Fix**: Route `BEAST_SOVEREIGN`, `PRIMORDIAL_SHIFTER`, `SAGE_SUPPORT`, `SONIC_HARMONY`, `SPIRIT_HUSBANDRY`, and `TIDE_VOYAGER` leaders to `'SANCTUM_MYRIAD_BEAST_TOTEM'`, and weight Sanctum selection across all 4 Sanctums based on missing `CAT_*` catalysts and path resonance!

---

2. `[REALM-MIC-02]` Unfixed `OUT-05` & `OUT-18`: Zero-Pool Stone Minting & Instant `0`-Tick Resolution in Micro-Locations & Secret Realm Races!

1. **`OUT-05` — `EX.dispatchExploration` Mints `+820` Stones When `W.world.resourcePool.stones === 0` & Uses `0` Travel Time (`js/explore.js:8299–8405`, Verified in `outside_probe.js:11`)**:
   - **Exact Bug**: `W.world.resourcePool.stones = Math.max(0, pool - reward)` clamps the pool at `0` while **`ECO.earn(W, reward)` still credits the full `reward` (`+820` stones across the 4 spots!)**, violating Closed-Loop conservation! Furthermore, `day`/`t` do not advance, `m.away` is never set, and **zero autonomous NPCs ever visit the 4 Micro-Locations**!
   - **Exact Fix**:
     - Strictly compute `const actualReward = (W.world && W.world.resourcePool) ? Math.min(rawReward, Math.max(0, W.world.resourcePool.stones)) : rawReward;` and deduct/credit `actualReward` identically!
     - Wire autonomous pilgrimages (`AI` / `EXPED` `1`-day short expeditions when `opts.autonomous === true`, keeping synchronous mode when `v400` calls `dispatchExploration` directly!):

    - Sword cultivators (`SWORD` / `XIAN`) $\to$ `LOC_NAMELESS_SWORD_TOMB` (`+SWORD` mastery & sword intent)
    - Puppet/Craft cultivators (`MECH_PUPPET` / `CRAFT_CREATION`) $\to$ `LOC_ANCIENT_MECH_RUINS` (`+MECH_GEARS` & blueprints)
    - Venom/Herb cultivators (`VENOM_GU` / `herbal`) $\to$ `LOC_MYRIAD_VENOM_FOREST` (`rare herbs & Gu`)
    - High-`heartDemon` cultivators (`heartDemon >= 42`) $\to$ `LOC_MOON_MIRROR_VALLEY` (`cleanses -18 heartDemon!`)
2. **`OUT-18` — Single-Click Continental Secret Realm Race (`ui.js:3039`, `expedition.js:5187–5440`)**:
    - `trigger-secret-realm` calls `spawnSecretRealm` and immediately calls `resolveSecretRealmRace` in the same synchronous callback (`0` travel ticks, auto-drafting top 3 idle members)!
    - **Fix**: Spawn Secret Realms in `'OPEN'` state (`window = 5 days`), allowing our `mission`/`commander` officer (and rival sects!) to assemble and march real parties to the portal before `resolveSecretRealmRace` executes on the closing day!

---

3. `[REALM-TOMB-03]` Zero-Party Auto-Win on 3-Layer Immortal Tombs (`SECT.IMMORTAL_TOMB`, `v360` Phase 1), Un-Debited Puzzle Gates (`v320` Phase 8), & Branching Node AI!

1. **`SECT.IMMORTAL_TOMB.excavateTombLayer` Requires `0` Disciples, `0` Time & Has `0%` Trap/Boss Failure Risk (`test/v360:62–72`)!**:
    - `excavateTombLayer(W, siteId)` clears Layer 1 (`Outer Gate Seal: ARRAY + FORGING`), Layer 2 (`Epigraph Chamber: SCRIPTURE`), and Layer 3 (`Tomb Guardian Boss`) without even taking a `party` parameter unless optional!
    - **Fix**: Preserve default fallback for `v360` unit calls, while adding autonomous tomb-raiding expeditions (`stage >= 5` party of 3–4 specialists: an `ARRAY`/`FORGING` breaker, a `SCRIPTURE`/`LORE_ARCHIVIST` scholar, and `vanguard` fighters) with real trap/guardian injury checks and Closed-Loop stone rewards from `W.world.resourcePool.stones`!
2. **`EXPLORE.solveElementalPuzzleGate` (`v320:167–178`, `explore.js:8135`)**:
    - **Closed-Loop Leak**: Credits `rewardStones` (`300–400` stones) without debiting `W.world.resourcePool.stones`!
    - **Elemental Detection**: Check not only `m.h.root.primaryElement`, but also **equipped elemental weapons, `socketedRunes` (`E05`), and active `m.pets` elements**, and normalize `'DARK_YIN'` $\leftrightarrow$ `'YIN_SHADOW'`!

3. **5-Stage Branching Secret Realm Nodes (`SECT.SECRET_REALM_NODES`) & On-Map Compound Ruins (`cave`, `stele`, `spring`, `ruin`)**:
    - Ensure party leaders in `SECRET_REALM_NODES` choose between Combat, Trap/Array, and Lore/Puzzle branches using their actual `Human Core caution` vs `risk` and party skills (`ARRAY`, `SCRIPTURE`, `MOVEMENT`).
    - Upgrade on-map `spring` (น้ำพุปราณลับ) so ascetics upholding `VOW_ABSTAIN_PILLS` (`06A`) actively bathe there to purge `c.toxic` and mend minor injuries without breaking their vow, and scale `stele` (แท่นศิลาจารึกโบราณ) comprehension insights into mid/late game!

---

4. Mandatory Verification Suite Additions (`08C`)

1. **43-Floor Pagoda & 4th Sanctum Autonomous Reachability Test**: Give a Stage-7 `BEAST_SOVEREIGN` party `EX.tryVolunteer` opportunities and assert both `SANCTUM_MYRIAD_BEAST_TOTEM` and Pagoda Floors `>= 10` are entered!
2. **Closed-Loop `resourcePool === 0` Micro-Location & Puzzle Gate Test (`OUT-05`)**: Set `W.world.resourcePool.stones = 0`, run `EX.dispatchExploration` and `solveElementalPuzzleGate`, and assert `W.sect.stones` gains `0` unbacked Spirit Stones.
3. **Staged Secret Realm Race Test (`OUT-18`)**: Spawn a Secret Realm in `'OPEN'` state, advance days for party travel, and verify multi-sect race resolution occurs at portal closing.